Investigation of Current-Phase Relationship in 1T-Phase Platinum Ditelluride Josephson Junctions under In-Plane Magnetic Field

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Introduction

Outline

Introduction

- Research motivation
- 1T-$PtTe_2$ (Type-II Dirac semimetal)

Theory

- Josephson effect & CPR
- Josephson diode effect (JDE)

Methods

- Device fabrication
- Measurement setup

Theoretical Background

Josephson Effect & Current–Phase Relation

Josephson relations

- DC: $I_s = I_c \sin \phi$ (ideal tunnel junction)
- AC: $\dot{\phi} = \frac{2e}{\hbar} V$

Beyond sinusoidal CPR

- High transparency → **skewed** CPR
- Harmonics: $I(\phi) = \sum_n I_n \sin(n\phi + \delta_n)$
- Access via **asymmetric SQUID** readout

Josephson Diode Effect (Concept)

Definition

- Non□reciprocal critical current

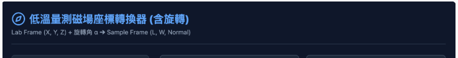
Experimental Methods

Device Fabrication Flow

1. **Crystal Growth**: Self-flux method (provided by Prof. C.S. Lue, NCKU).
2. **Exfoliation**: Mechanical exfoliation of $PtTe_2$ flakes onto SiO_2/Si substrates.
3. **E-beam Lithography (EBL)**:
   - PMMA resist coating.
   - Pattern definition for contacts and SQUID loops.
4. **Metallization**:
   - Superconducting electrodes: **Al (Aluminum)** or **Nb (Niobium)**.
   - *In-situ* Ar ion milling to ensure transparent interfaces.
   - E-beam evaporation & Liftoff.
5. **Design**: Asymmetric dc-SQUID geometry for CPR retrieval.

Measurement Setup

- **Cryogenic system**: Dilution refrigerator (base ~40 mK)



Results: Basic Characterization

I-V Characteristics (DC Transport)

- **Typical IV Curve**:
  - Clear Superconducting branch ($V = 0$).
  - Switching Current (I_c) and Retrapping Current (I_r).
  - Hysteretic behavior (underdamped junction).

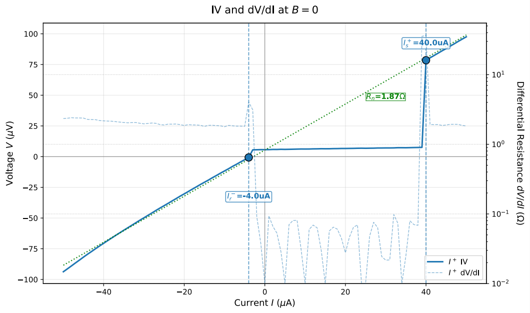


Figure 3: I-V Curve at Base Temp.

- **Observation**:
  - Sharp transition indicates high-quality junction.
  - Finite resistance in normal state (R_N).
  - **Excess Current** observed at high bias $\rightarrow$ High transparency interface.

Results: Current-Phase Relation

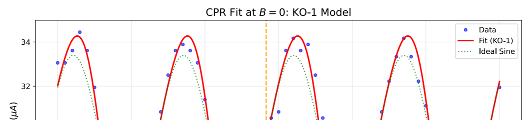
CPR Retrieval Method

- **Asymmetric SQUID Technique**:
  - Large junction (I_{ref}) // Small junction (I_{test}).
  - $I_c(\Phi_{ext}) \approx I_{ref} + I_{test}(\phi)$.
  - Assumes $I_{ref} \gg I_{test}$ and sinusoidal $I_{ref}(\phi)$.
- **Math Model**:

$$I_s(\phi) = I_1 \sin(\phi) + I_2 \sin(2\phi + \delta)$$

  - I_1: Fundamental component (conventional).
  - I_2: Second harmonic (4π-periodic / ABS).

Measured CPR & Fitting

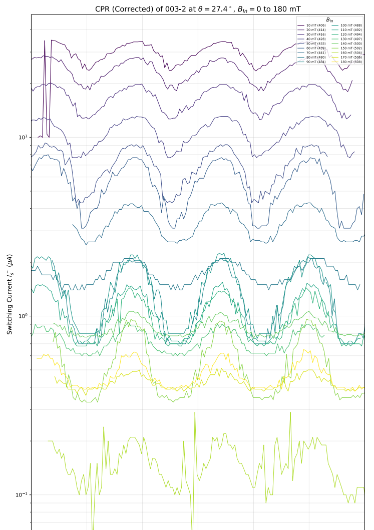


- **Key Findings**:
  - **Skewed CPR**: Deviation from pure sine wave.
  - **High I_2/I_1 ratio**: Significant

Results: In-Plane Magnetic Field Effect

Tunable CPR with In-Plane Field

- Applying $B_{\parallel}$ **modifies CPR shape**
- Suggests a **phase shift** (ϕ_0) in higher harmonics
- Correlates with emergence of the **Josephson diode effect**



Conclusion

Summary

1. **Successful Fabrication**:
    - High-quality 1T-$PtTe_2$ Josephson junctions and SQUIDs realized.
2. **Anomalous CPR**:
    - Observed significant **second harmonic** (I_2) contribution.
    - Direct evidence of high-transparency transport modes.
3. **Magnetic Tuning**:
    - In-plane magnetic field effectively modulates the CPR and induces **Josephson Diode Effect**.
    - Demonstrates potential for **tunable superconducting logic** and **phase batteries**.

Future Outlook

- **Optimization**: Improving fabrication yield and interface control.
- **Material Exploration**: Extending to other Weyl/Dirac semimetals.
- **Quantum Information**: Integration into superconducting qubit circuits (e.g., as

References